BIOL6130 Final Project: data preparation

Pilot study in area B

The initial study used a 4 km by 4 km window to inspect the land-cover data and one main road. The regional maps and environmental table follow this pilot.

Read the data

```
library(sf)
library(terra)
library(dplyr)
library(ggplot2)
library(tidyterra)

photo_2018 <- rast("Raw Data/dgt_ortho2018_context.tif")
photo_2025 <- rast("Raw Data/dgt_ortho2025_context.tif")
land_cover <- read_sf("Raw Data/cos2018_land_cover.geojson")
forest_2018 <- read_sf("Raw Data/cos2018_eucalyptus.geojson")
forest_2025 <- read_sf("Raw Data/cos2025_eucalyptus.geojson")
road <- read_sf("Raw Data/cos2018_roads.geojson") %>%
  filter(ogc_fid == 147070)

# Crop climate data to the current B window.
area <- as.polygons(ext(photo_2018), crs = crs(photo_2018))
window <- project(area, "EPSG:4326")
climate_file <- "Raw Data/PRT_wc2.1_30s_bio.tif"
climate_url <- paste0(
  "https://geodata.ucdavis.edu/climate/worldclim/2_1/tiles/iso/",
  "PRT_wc2.1_30s_bio.tif")
if (!file.exists(climate_file)) {
  download.file(climate_url, climate_file, mode = "wb", quiet = TRUE)
}
climate <- rast(climate_file)[[c(1, 12)]] %>%
  crop(window, snap = "out")
names(climate) <- c("temperature", "precipitation")
```

Inspect the contents

```r
glimpse(land_cover, width = 75)
```

```
Rows: 6,063
Columns: 12
$ ogc_fid      <int> 1288, 1294, 1296, 1297, 1299, 1323, 1361, 1367, 1368~
$ cos18v4_n4_c <chr> "1.1.1.1", "1.1.1.1", "1.1.1.1", "1.1.1.1", "1.1.1.1~
$ cos18v4_n4_l <chr> "Áreas edificadas residenciais contínuas predominant~
$ id           <int> 1288, 1294, 1296, 1297, 1299, 1323, 1361, 1367, 1368~
$ cos18v4_n3_c <chr> "111", "111", "111", "111", "111", "111", "111", "11~
$ cos18v4_n3_l <chr> "Áreas edificadas residenciais contínuas", "Áreas ed~
$ cos18v4_n2_c <chr> "11", "11", "11", "11", "11", "11", "11", "11", "11"~
$ cos18v4_n2_l <chr> "Áreas edificadas residenciais", "Áreas edificadas r~
$ cos18v4_n1_c <chr> "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1~
$ cos18v4_n1_l <chr> "Territórios artificializados", "Territórios artific~
$ area_ha      <dbl> 2.5112068, 4.9795026, 1.3266062, 7.0387354, 2.877182~
$ geometry     <MULTIPOLYGON [°]> MULTIPOLYGON (((-8.251433 4..., MULTIPO~
```

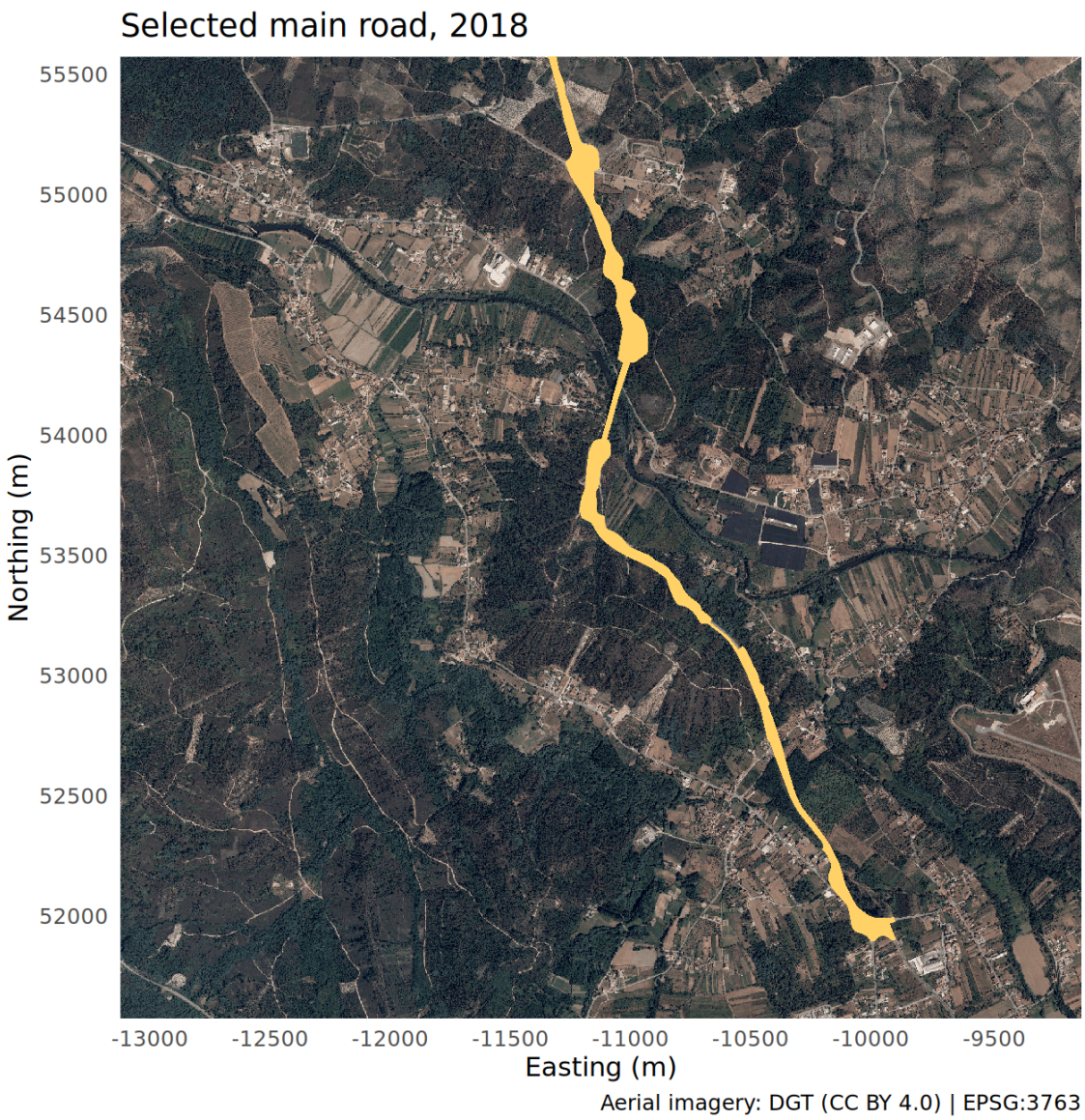
Draw the pilot maps

The pilot focused on the main road recorded as COS ID 147070. The yellow overlay marks this road within area B.

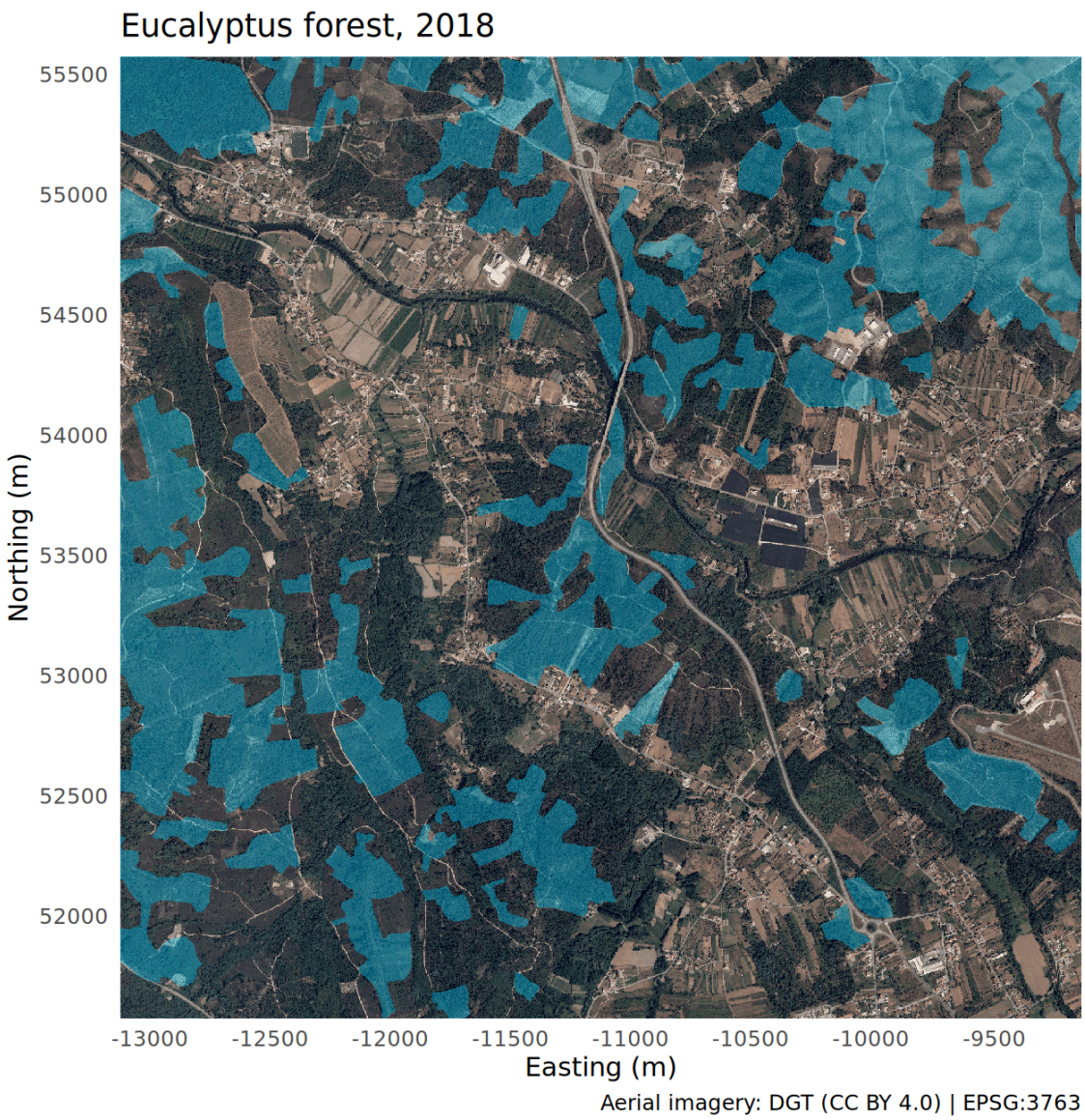
The climate maps show 1970–2000 averages at 30-arc-second resolution (about 710 by 925 m here). The 2025 aerial image provides geographic context.

```r
# tidyterra draws the photograph; ggplot2 adds the data layer and map layout.
draw_map <- function(photo, layer, title) {
  ggplot() +
    geom_spatraster_rgb(data = photo, interpolate = FALSE,
                        maxcell = ncell(photo)) +
    layer +
    coord_sf(crs = 3763, datum = st_crs(3763), expand = FALSE,
             xlim = c(xmin(photo), xmax(photo)),
             ylim = c(ymin(photo), ymax(photo))) +
    labs(title = title, x = "Easting (m)", y = "Northing (m)",
         caption = "Aerial imagery: DGT (CC BY 4.0) | EPSG:3763") +
    theme_minimal()
}
```

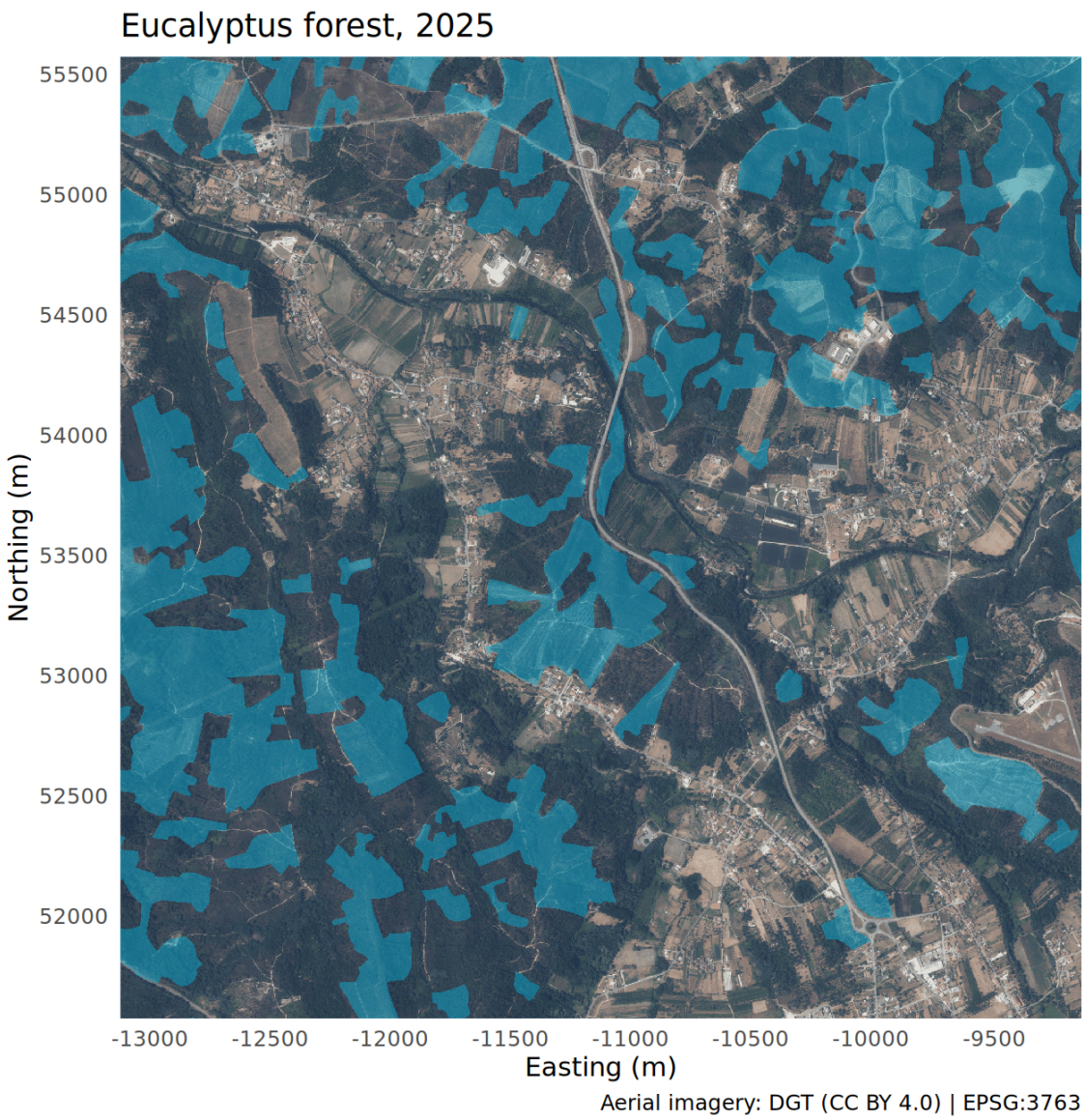
```r
draw_map(photo_2018,
         geom_sf(data = road, fill = "#FFD166", colour = NA),
         "Selected main road, 2018")
```



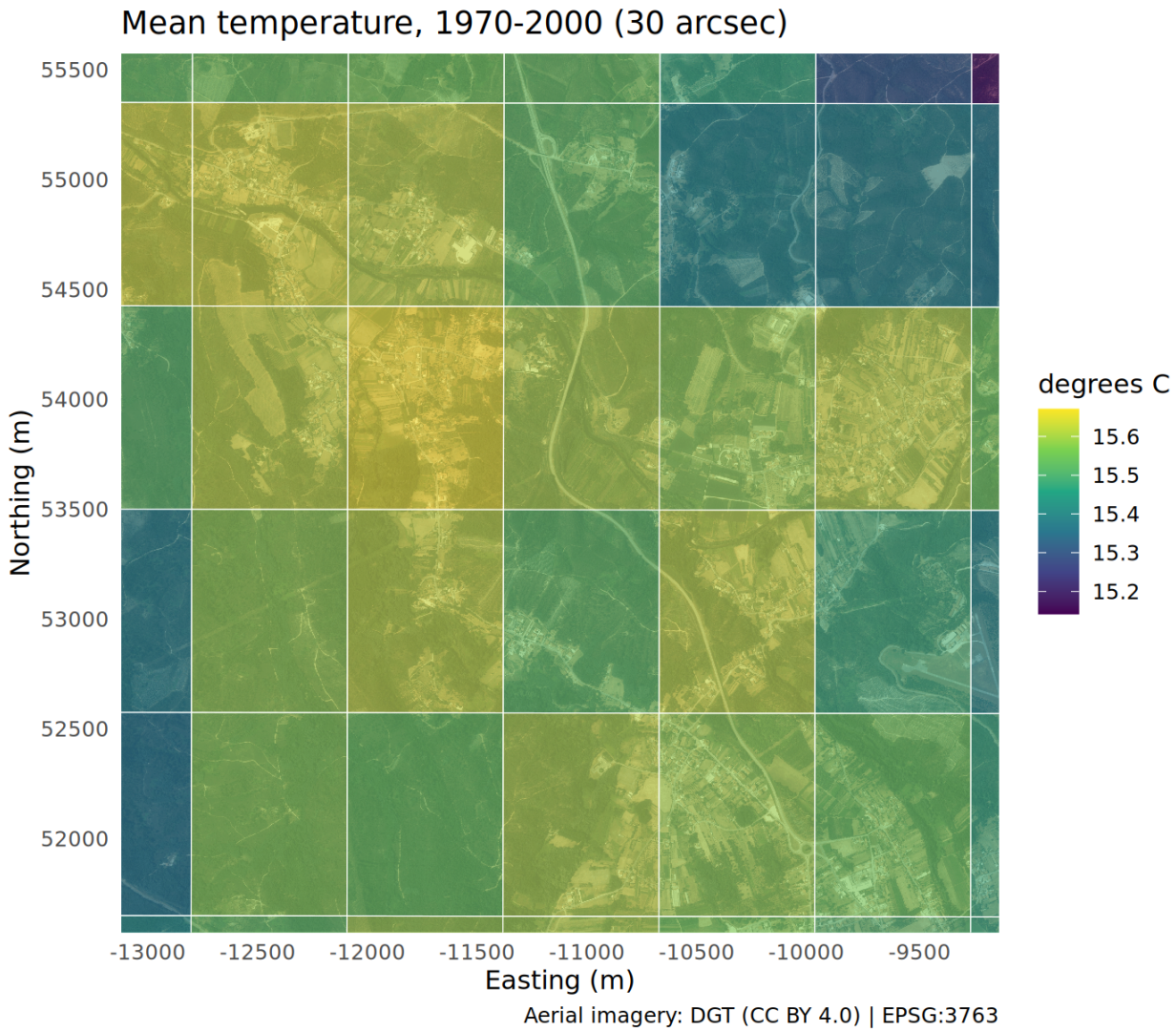

```
draw_map(photo_2018,
         geom_sf(data = forest_2018, fill = "#00BFE8",
                 colour = NA, alpha = 0.4),
         "Eucalyptus forest, 2018")
```



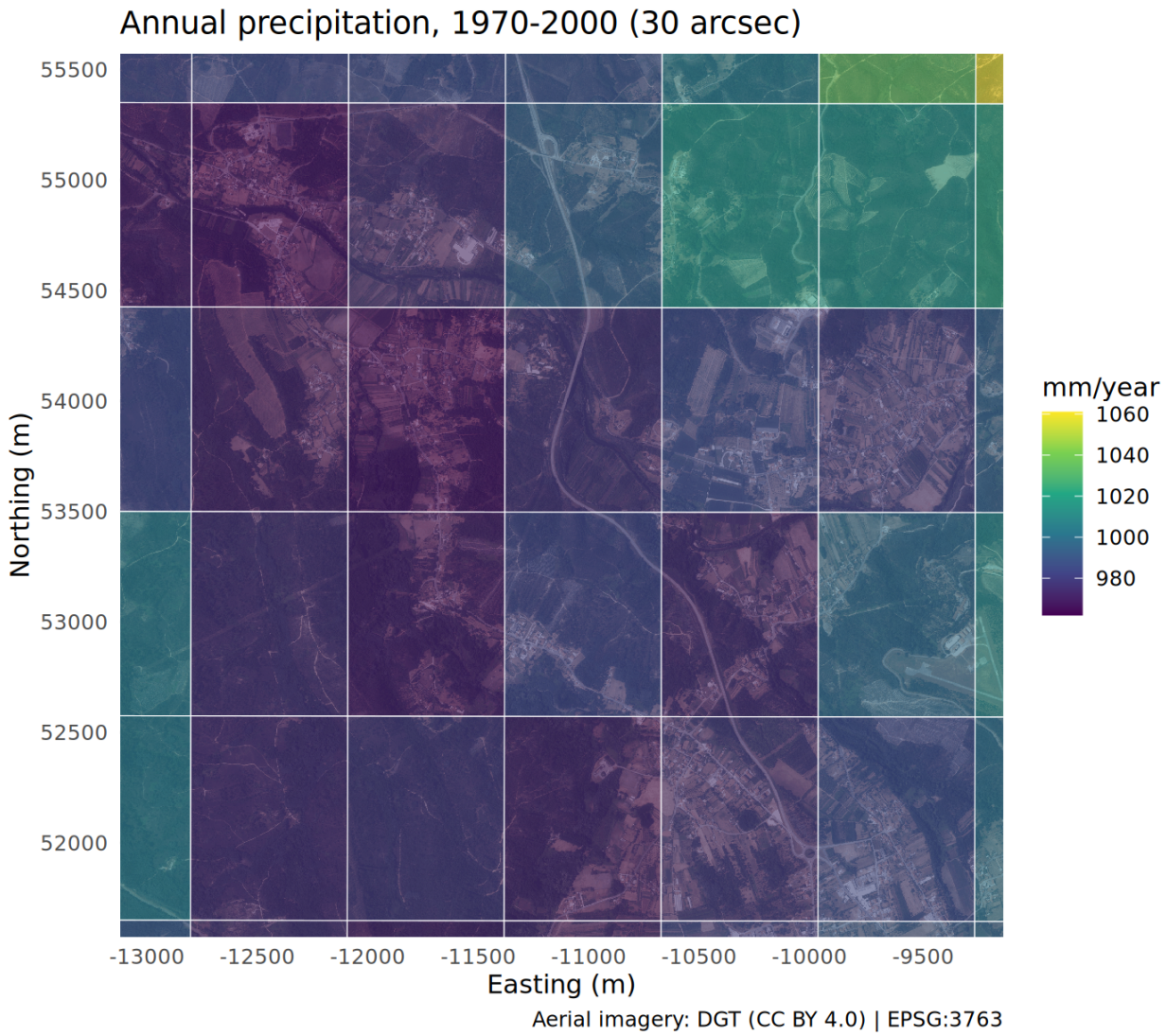

```
draw_map(photo_2025,
         geom_sf(data = forest_2025, fill = "#00BFE8",
                 colour = NA, alpha = 0.4),
         "Eucalyptus forest, 2025")
```



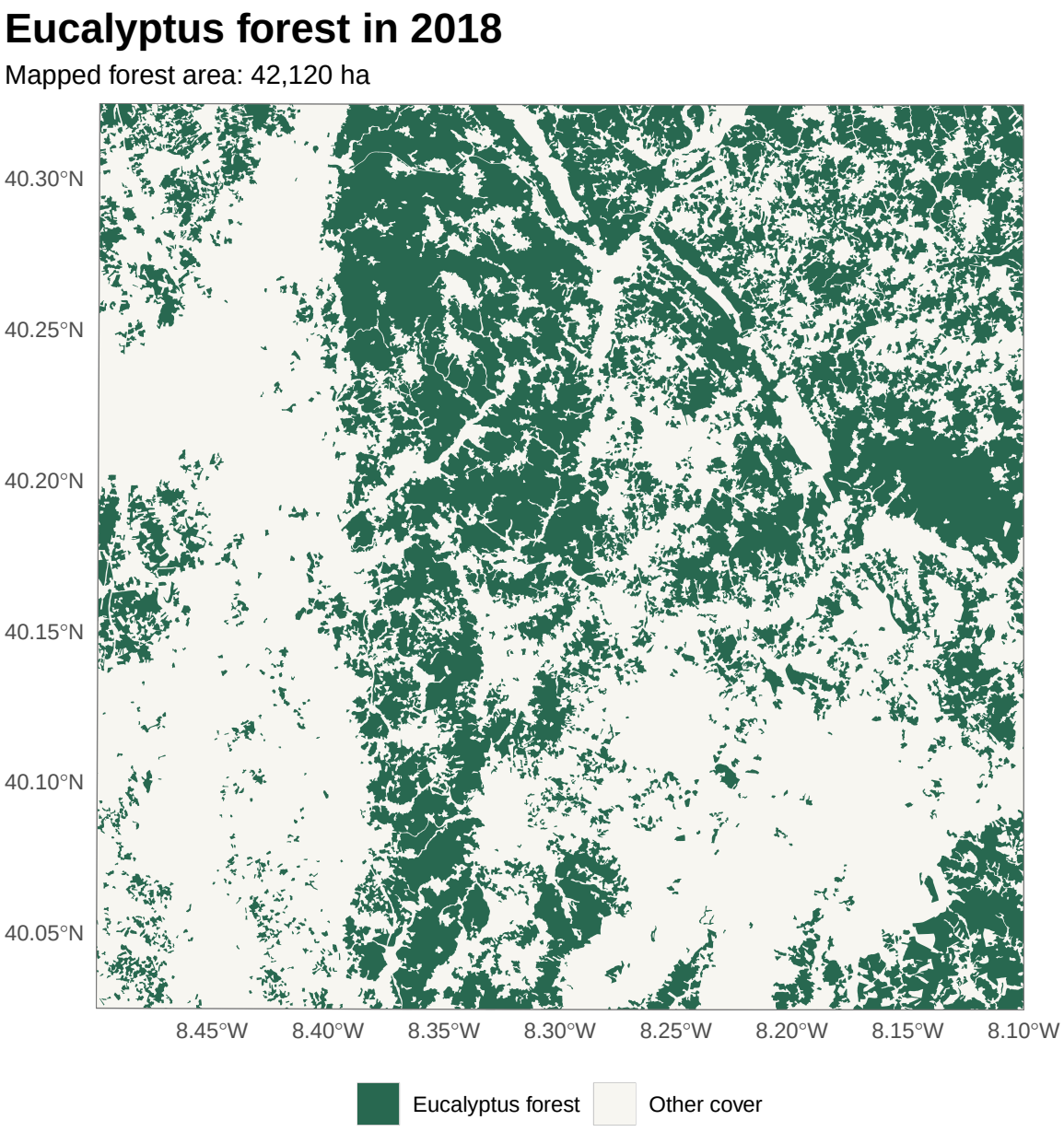

```
# Keep the original climate cells and values.
cells <- as.polygons(climate, aggregate = FALSE, round = FALSE) %>%
  st_as_sf()
draw_map(photo_2025,
         geom_sf(data = cells, aes(fill = temperature),
                 colour = "white", alpha = 0.55),
         "Mean temperature, 1970-2000 (30 arcsec)") +
  scale_fill_viridis_c(name = "degrees C")
```



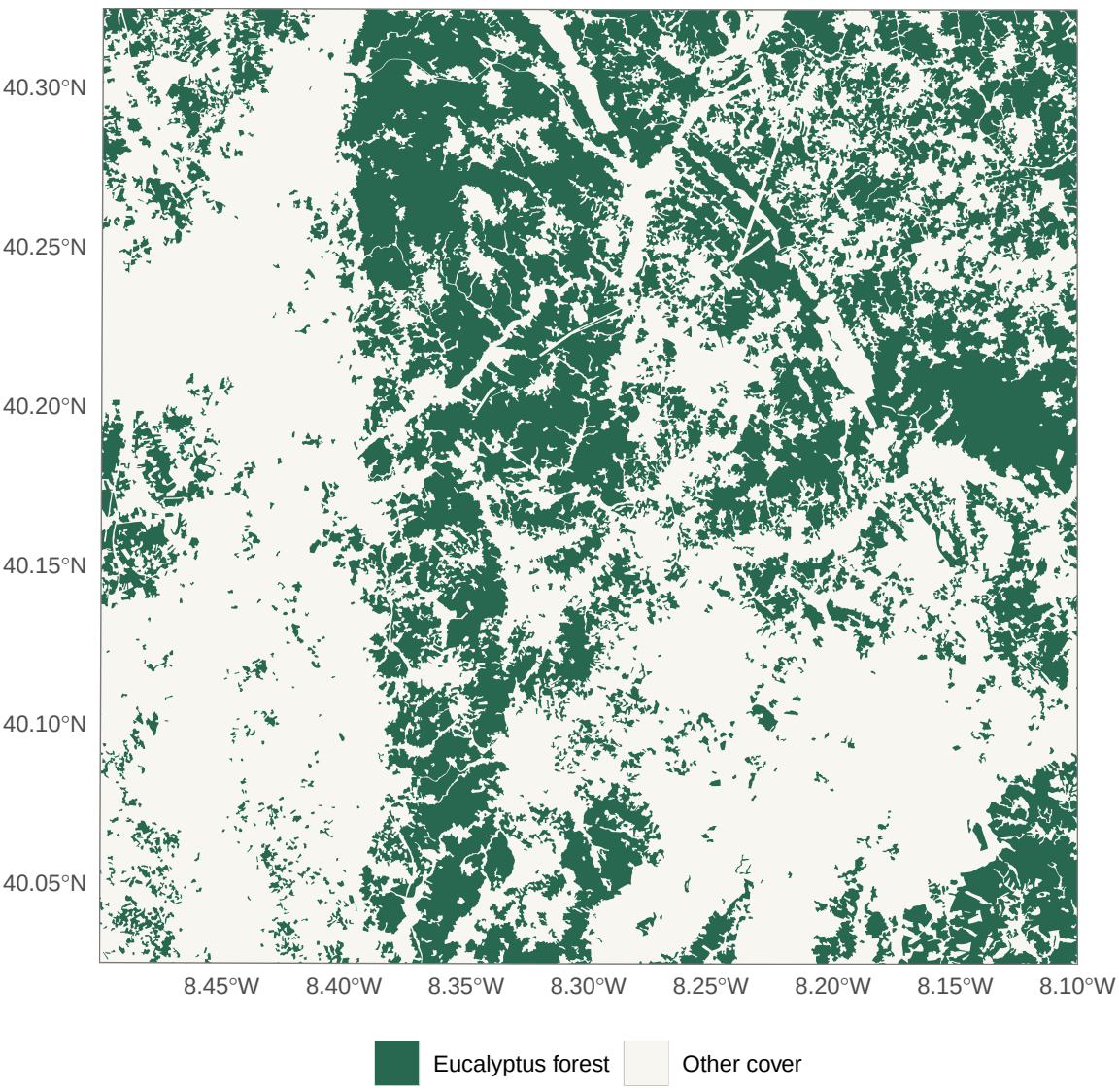

```
draw_map(photo_2025,
         geom_sf(data = cells, aes(fill = precipitation),
                 colour = "white", alpha = 0.55),
         "Annual precipitation, 1970-2000 (30 arcsec)") +
  scale_fill_viridis_c(name = "mm/year")
```


Annual precipitation, 1970-2000 (30 arcsec)
55500
55000
54500
54000
53500
53000
52500
52000
Northing (m)
-13000
-12500
-12000
-11500
-11000
-10500
-10000
-9500
Easting (m)
mm/year
1060
1040
1020
1000
980
Aerial imagery: DGT (CC BY 4.0) | EPSG:3763

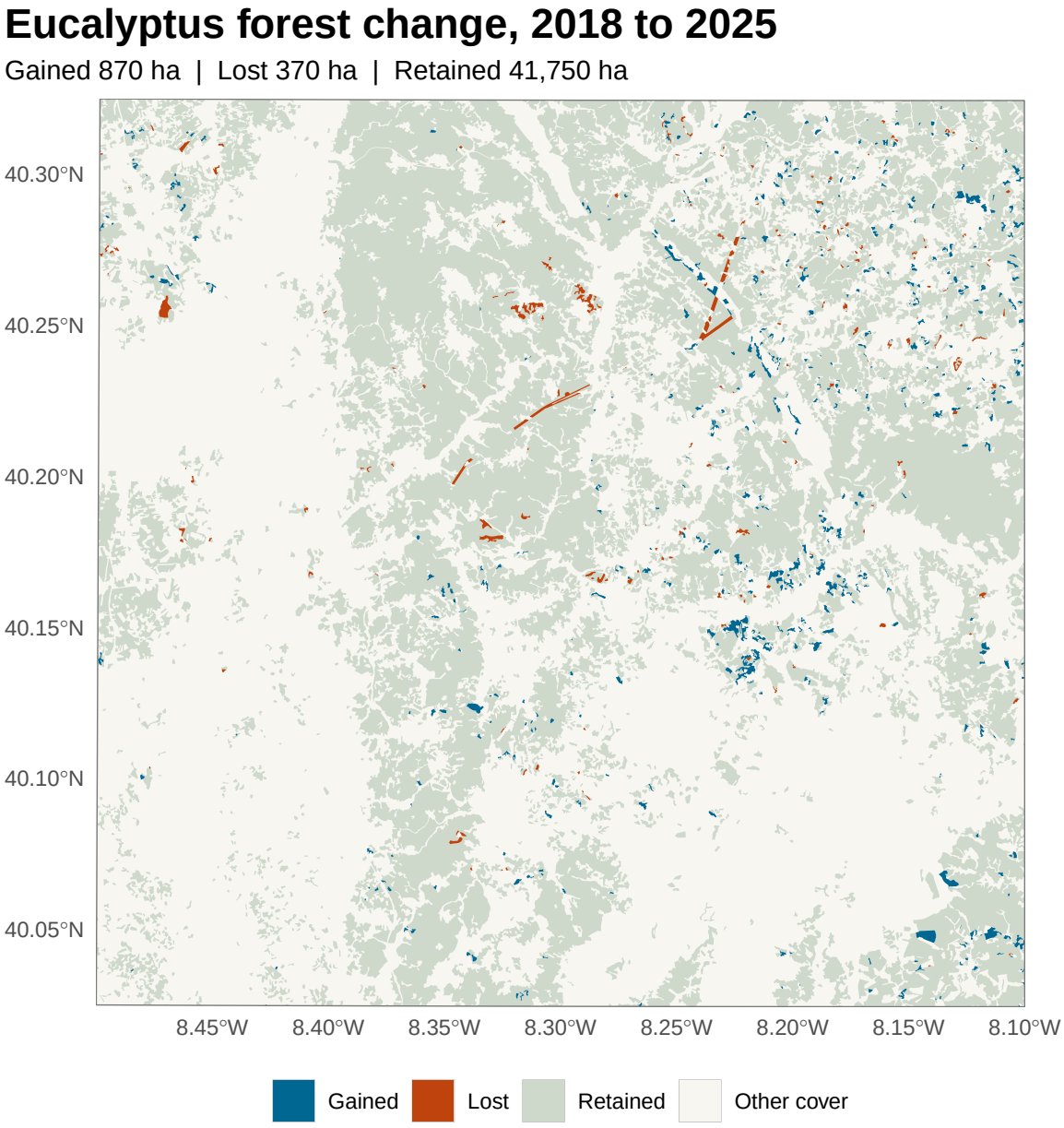
Regional eucalyptus forest cover



The map covers the full study window near Coimbra, central Portugal (8.50–8.10°W and 40.025–40.325°N). Green shows every polygon classified as eucalyptus-dominated forest in the 2018 Portuguese land-cover map.



The 2025 map uses the same boundary, scale and eucalyptus class as the 2018 map. Each annual area is measured from the original land-cover polygons clipped to that boundary.



Blue marks land mapped as eucalyptus only in 2025. Orange marks land mapped as eucalyptus only in 2018. Pale green marks areas mapped as eucalyptus in both years. The net gain was **500.3 ha**. The map records a change in the dominant cover class. Individual seedlings, planting dates and routes of seed arrival cannot be identified from these polygons.

Environment table

Each row in `environment.csv` is one point on the 80 m grid. It brings together temperature, rainfall, wind speed, elevation, slope, land cover and fire and forest management records. We can use these columns later to find places with similar conditions near and far from roads. Nearby points often share the same climate cell.

Data sources

- **Land cover and the pilot road.** Direção-Geral do Território (DGT), COS Series 2, 2018 and 2025. The forest maps use eucalyptus class 5.1.1.6. DGT, CC BY 4.0
- **Pilot aerial photos.** DGT imagery from 2018 and 2025, exported at 2 m for area B. DGT orthophotos, CC BY 4.0
- **Pilot climate.** WorldClim v2.1, temperature and rainfall averages for 1970–2000. WorldClim · Fick and Hijmans (2017)
- **Regional climate.** CHELSA v2.1, temperature, rainfall and mean near-surface wind speed for 1981–2010. CHELSA · BIOCLIM+ data, CC0
- **Elevation and slope.** Copernicus DEM GLO-30, tile N40/W009. Product and licence
- **Fire.** ICNF, *Áreas ardidas*, 2017–2025, retrieved 5 November 2026. Fire perimeters · ICNF terms. The 2017 burn severity and image quality layers come from Jahanianfard et al. Portuguese Burn Severity Atlas v2, CC BY 4.0
- **Forest management.** Lesiv et al., global forest management map for 2015, v3.2. Data, CC BY 4.0

Run `quarto render analysis.qmd` to rebuild the figures and CSV. The first run downloads the WorldClim file for the pilot maps if needed.

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